

2. Arithmetization (VM circuits)

1. Execution layer

guest program
(PLUM Verify / BDEC)

Executor (rv32im)
emit execution trace

execution trace
(clk, pc, instr, . . .) rows

3. Proof layer

commit
(Merkle / FRI)

PIOP
sumcheck/STIR

recursion
+ wrap

receipt

memory chip

CPU chip

precompile chips
Griffin- $\mathbb{F}_p^{192}$ (this work)
+ bigint, byte-range

lookup

*Field-mismatch tax arises here: PLUM's $\mathbb{F}_p^{199}$ ops over the 31-bit prover field.
The Griffin precompile (orange) absorbs the dominant share; layered taxonomy after the zkVM SoK (ePrint 2026/525).*